

Since you're aiming to **design and innovate a McKibben artificial muscle sheath** (not just understand how it works), you should study books from **mechanics, composite materials, pressure vessels, soft robotics, biomechanics, and finite element analysis**. There is **no single textbook dedicated entirely to McKibben muscles**, so researchers build knowledge from several disciplines.

1. Soft Robotics (Highest Priority)

Soft Robotics: Trends, Applications and Challenges

Editors: Cecilia Laschi, Barbara Mazzolai, Matteo Cianchetti

Why study it?

- Soft actuator fundamentals
- Pneumatic artificial muscles
- Bio-inspired actuator design
- Design methodologies

Recommended chapters

- Soft actuators
- Pneumatic actuators
- Artificial muscles

Soft Robotics: From Biological Inspiration to Engineering Design

Authors: Cecilia Laschi & Barbara Mazzolai

Why?

- Mathematical modeling
- Soft actuator mechanics
- Design concepts
- Material selection

★ ★ ★ ★ ★ (Highly recommended)

2. Continuum Mechanics

Introduction to Continuum Mechanics

Author: W. Michael Lai

Study

- Stress tensor
- Strain tensor
- Pressure loading
- Deformation
- Equilibrium

Essential for deriving McKibben equations.

3. Mechanics of Materials

Mechanics of Materials

Author: R. C. Hibbeler

Study

- Axial loading
- Torsion
- Stress concentration
- Fatigue
- Elastic deformation

Useful for sheath fibers and end fittings.

Mechanics of Materials

Authors: Beer, Johnston, DeWolf

Also excellent.

4. Pressure Vessel Design

Pressure Vessel Design Manual

Author: Dennis Moss

Study

- Thin cylinder theory
- Hoop stress
- Axial stress
- Internal pressure
- End cap loading

This explains why the bladder expands and why the sheath is necessary.

5. Composite Materials

Mechanics of Composite Materials

Author: Robert M. Jones

Very important because the braided sheath behaves like a braided composite.

Study

- Fiber orientation
 - Orthotropic materials
 - Fiber stress
 - Composite stiffness
-

Engineering Mechanics of Composite Materials

Author: Isaac Daniel

Excellent for

- Fiber architecture
- Braided composites
- Textile mechanics

6. Textile Reinforcement

Braided Structures and Composites

Author: Yordan Kyosev

This is probably the **best book** for your sheath research.

Study

- Braiding theory
- Diamond braid geometry
- Braid angle
- Carrier design
- Fiber interaction

★ ★ ★ ★ ★ (One of the most important references)

Textile Structural Composites

Author: T. W. Chou

Study

- Braided textiles
- Woven reinforcement
- Knitted reinforcement
- Fiber architecture

7. Contact Mechanics

Contact Mechanics

Author: K. L. Johnson

Study

- Fiber contact

- Friction
- Hertzian contact
- Sliding
- Wear

Very useful if you want to improve the crossover design of the braid.

8. Finite Element Analysis

Concepts and Applications of Finite Element Analysis

Author: Robert D. Cook

Study

- Large deformation
- Hyperelasticity
- Nonlinear analysis
- Contact
- Stress analysis

Essential if you plan to validate your design in ANSYS.

9. Hyperelastic Materials

Nonlinear Solid Mechanics

Authors: Gerhard Holzapfel

Study

- Rubber mechanics
- Neo-Hookean model
- Mooney-Rivlin model
- Ogden model

Very useful because the bladder is hyperelastic.

10. Biomechanics

Biomechanics: Mechanical Properties of Living Tissues

Author: Y. C. Fung

Study

- Muscle mechanics
- Soft tissues
- Stress-strain behavior

This helps you understand the biological inspiration behind artificial muscles.

McKibben-Specific Papers (Must Read)

These are considered foundational references:

1. **Joseph L. McKibben** – Original Artificial Muscle Patent (1950s)
2. **Schulte (1961)** – *The Characteristics of the McKibben Artificial Muscle*
3. **Chou & Hannaford (1996)** – Measurement and Modeling of McKibben Muscles
4. **Tondu & Lopez (2000)** – Modeling and Control of McKibben Artificial Muscle Robot Actuators
5. **Daerden & Lefeber (2002)** – Pneumatic Artificial Muscles: Actuators for Robotics and Automation

These papers are cited by almost every modern McKibben muscle publication.

If your research is specifically on the sheath

Focus on these subjects in order:

1. **Braided Structures and Composites** – Yordan Kyosev ★★★★★
2. **Mechanics of Composite Materials** – Robert M. Jones ★★★★★
3. **Pressure Vessel Design Manual** – Dennis Moss ★★★★★☆
4. **Contact Mechanics** – K. L. Johnson ★★★★★☆
5. **Soft Robotics: From Biological Inspiration to Engineering Design** – Laschi & Mazzolai ★★★★★
6. **Nonlinear Solid Mechanics** – Gerhard Holzapfel ★★★★★☆

This combination will give you the theoretical foundation to design, model, analyze, and optimize a novel McKibben sheath, covering geometry, fiber mechanics, pressure loading, contact behavior, materials, and soft robotic applications.

If you are **designing a completely new McKibben sheath**, don't think only about the braid angle. Researchers usually consider **multiple design parameters simultaneously**. You can group them into geometric, material, mechanical, manufacturing, and performance considerations.

Design Considerations for a McKibben Sheath

1. Geometrical Parameters (Most Important)

These define the kinematics of the actuator.

(a) Initial Braid Angle (θ)

- Controls contraction ratio
- Controls output force
- Controls actuator stiffness
- Determines whether the actuator contracts or extends
- Affects efficiency

Design question:

What initial braid angle provides the required force and stroke?

(b) Fiber Orientation

Possible arrangements include:

- Diamond braid (traditional)
- Variable-angle braid
- Hybrid braid
- Spiral
- Knitted
- Woven
- Lattice

Design question:

Which fiber architecture provides the desired motion?

(c) Braid Pitch

Distance between two consecutive turns.

Affects

- flexibility
- stress distribution
- contraction
- stiffness

(d) Number of Helical Turns

More turns

↓

Higher flexibility

↓

Lower local stress

But

↓

More friction

(e) Fiber Density

Determines

- open area
- load sharing
- radial stiffness
- contraction

Higher density

- higher strength
 - lower expansion
 - heavier sheath
-

(f) Diamond Cell Geometry

Consider

- diamond length
- diamond width
- aspect ratio
- opening size

These influence

- pressure distribution
 - deformation
 - local stresses
-

2. Material Selection

The fiber material should satisfy

- High tensile strength
- Low elongation
- High fatigue resistance
- Low creep
- High flexibility
- Abrasion resistance

Possible materials

- PET
- Nylon
- Kevlar
- Vectran
- Dyneema
- Carbon fiber (limited flexibility)
- Glass fiber (limited flexibility)

3. Mechanical Considerations

(a) Fiber Tension

Each fiber experiences tensile loading.

Questions

- Maximum tension?
- Uniform distribution?
- Stress concentration?

(b) Radial Constraint

The sheath should

- allow controlled expansion
- prevent ballooning
- avoid excessive radial deformation

(c) Axial Force

The sheath must efficiently convert

Pressure

↓

Axial force

Questions

- Maximum force?
 - Force efficiency?
 - Force stability?
-

(d) Contraction Ratio

Typical values

20–35%

Question

Can the design increase contraction while maintaining force?

(e) Radial Stiffness

Higher stiffness

↓

Less expansion

↓

Higher force

Lower stiffness

↓

Higher stroke

↓

Less force

(f) Torsional Stability

The actuator should not twist unless twisting is desired.

Consider

- balanced braid
- equal fiber angles

- symmetric structure
-

4. Contact Mechanics

Often ignored in basic designs but very important.

Fiber-to-Fiber Friction

Causes

- hysteresis
 - heat
 - wear
-

Contact Pressure

Higher contact pressure

↓

Higher friction

↓

Higher energy loss

Sliding Between Fibers

Influences

- efficiency
 - fatigue
 - repeatability
-

5. Bladder–Sheath Interaction

The sheath should

- prevent bladder extrusion
 - maintain uniform contact
 - avoid stress concentration
 - distribute pressure evenly
-

6. Manufacturing Considerations

Consider

Number of Carriers

16

24

32

48

64

Manufacturing Method

- Textile braiding
 - Knitting
 - Weaving
 - 3D printing
 - Composite winding
-

Fiber Pretension

Loose braid

↓

Poor control

Tight braid

↓

Reduced contraction

An optimum pretension is required.

End Attachment

Very important.

Poor end fittings cause

- fiber pull-out
 - stress concentration
 - leakage
 - premature failure
-

7. Dynamic Performance

The sheath affects

- response time
- damping
- vibration
- natural frequency

Questions

Can the actuator operate at

- 1 Hz?
- 5 Hz?
- 10 Hz?

8. Fatigue Design

Consider

- cyclic loading
- creep
- abrasion
- fiber breakage
- coating degradation

Target

More than one million cycles

9. Failure Modes

A good design should avoid

- Fiber rupture
 - Braid opening
 - Ballooning
 - End pull-out
 - Bladder rupture
 - Local buckling
 - Uneven contraction
 - Permanent deformation
 - Twisting
 - Wear at crossover points
-

10. Performance Parameters

Evaluate

- Output force
- Stroke

- Efficiency
 - Hysteresis
 - Energy consumption
 - Specific force (N/kg)
 - Specific work (J/kg)
 - Repeatability
 - Reliability
-

11. Optimization Parameters

Modern researchers optimize

- Braid angle
 - Fiber diameter
 - Fiber spacing
 - Carrier number
 - Material combination
 - Layer thickness
 - Surface coating
 - Cell geometry
 - Gradient stiffness
 - Variable braid angle
-

12. Research Considerations for a Novel Sheath

If your goal is to propose a **new sheath architecture** (rather than just changing the braid angle), the following aspects deserve particular attention because they are less explored in the literature:

Design Aspect	Why it Matters	Research Potential
Fiber architecture	Determines motion (contraction, bending, twisting)	★★★★★
Variable stiffness	Adapts to different loads	★★★★★

Design Aspect	Why it Matters	Research Potential
Contact mechanics	Reduces hysteresis and wear	★★★★★
Fiber tension distribution	Prevents localized failure	★★★★☆
Surface friction control	Improves efficiency	★★★★☆
Multi-material reinforcement	Balances strength and flexibility	★★★★☆
Graded braid geometry	Tailors local deformation	★★★★★
Smart fibers (sensing)	Enables self-monitoring	★★★★☆
Manufacturability	Reduces cost and improves repeatability	★★★☆☆
Fatigue life	Critical for long-term operation	★★★★★

A comprehensive design framework

A rigorous McKibben sheath design can be organized into five engineering domains:

1. **Geometric Design** – braid angle, pitch, fiber orientation, cell geometry.
2. **Material Design** – fiber material, coatings, stiffness, fatigue properties.
3. **Mechanical Design** – force transmission, tension distribution, radial constraint, torsional balance.
4. **Manufacturing Design** – fabrication method, pretension, end fittings, dimensional tolerances.
5. **Performance Design** – contraction ratio, efficiency, hysteresis, durability, repeatability, and dynamic response.

This framework is broad enough to guide the design of both conventional diamond-braided sheaths and novel reinforcement architectures, and it provides a logical structure for a research methodology chapter or a design optimization study.

If you are doing research, you should understand the **theoretical relationships** from continuum mechanics, pressure vessel theory, and helical geometry, rather than just memorizing the force equation. The classical theories were developed by Gaylord (1958), Schulte (1961), Chou & Hannaford (1996), and later refined by Tondu & Lopez (2000).

1. Pressure Vessel Theory

The McKibben muscle is essentially a **thin-walled cylindrical pressure vessel** covered by an inextensible braided sheath.

When internal pressure (P) is applied, it creates two principal stresses in the bladder.

Hoop (Circumferential) Stress

$$\sigma_h = \frac{Pr}{t}$$

where

- (P) = internal pressure
- (r) = bladder radius
- (t) = bladder thickness

Hoop stress is twice as large as axial stress.

Axial Stress

$$\sigma_a = \frac{Pr}{2t}$$

Thus,

$$\sigma_h = 2\sigma_a$$

This is why the bladder naturally wants to expand radially.

The sheath converts this radial tendency into axial contraction.

2. Helical Geometry Theory

Consider one braid fiber.

Its length is constant.

$$\boxed{b = \sqrt{L^2 + (2\pi nr)^2}}$$

where

- b = fiber length
- L = actuator length
- r = radius
- n = number of helical turns

Differentiate:

$$db = 0$$

Since

$$b = \text{constant}$$

Therefore

$$dL = -\frac{4\pi^2 n^2 r}{L} dr$$

Hence

$$dr > 0$$

implies

$$dL < 0$$

This proves mathematically that increasing diameter must reduce length.

This is the geometric origin of contraction.

3. Braid Angle Relation

The braid angle satisfies

$$\boxed{\tan\theta = \frac{2\pi nr}{L}}$$

Therefore

$$\boxed{\cos\theta = \frac{L}{b}}$$

and

$$\boxed{\sin\theta = \frac{2\pi nr}{b}}$$

These equations directly relate geometry to actuator behavior.

4. Fiber Tension Theory

Suppose each braid fiber carries tension

$$T$$

The fiber force can be resolved into components.

Axial component

$$\boxed{T_z = T\cos\theta}$$

Circumferential component

$$\boxed{T_\theta = T\sin\theta}$$

These two components explain almost every behavior of the actuator.

Small braid angle

$$\theta = 20^\circ \quad \cos 20 = 0.94$$

Nearly all fiber tension contributes to axial force.

Large braid angle

$$\theta = 70^\circ \quad \cos 70 = 0.34$$

Only a small part contributes to contraction.

5. Equilibrium of Forces

Pressure acts on the projected area

$$A = \frac{\pi D^2}{4}$$

Pressure force

$$\boxed{F_p = PA}$$

The fibers oppose this force.

Static equilibrium gives

$$\boxed{\sum F = 0}$$

which leads to the classical McKibben force equation.

6. Chou-Hannaford Force Model

Assuming

- constant fiber length
- negligible friction
- uniform pressure

the force is

$$\boxed{F = \frac{\pi D^2 P}{4} (3 \cos^2 \theta - 1)}$$

Observe

Pressure increases

↓

Force increases

Angle increases

↓

Force decreases

7. Critical Angle Theory

Set

$$F=0$$

Then

$$3 \cos^2 \theta - 1 = 0$$

Therefore

$$\boxed{\theta = 54.74^\circ}$$

At this angle

Axial force

=

0

This is called the **neutral angle**.

It is not the optimal design angle.

8. Why Does Force Reduce?

Pressure is radial.

The fiber converts radial force into axial force.

The conversion efficiency is

$$\boxed{\eta \propto \cos \theta}$$

Small angle

↓

large axial component

Large angle

↓

small axial component

9. Fiber Stress Theory

The tensile stress in one fiber is

$$\boxed{\sigma_f = \frac{T}{A_f}}$$

where

(A_f)

is the fiber area.

Higher pressure

↓

higher tension

↓

higher stress

↓

possible fiber failure

10. Strain Theory

Engineering strain

$$\epsilon = \frac{L - L_0}{L_0}$$

Negative strain

=

contraction

Theoretical contraction

≈36%

Practical

20–30%

because of

- friction
 - bladder stiffness
 - end effects
-

11. Energy Theory

Pressure work

$$W = P \Delta V$$

Output work

$$\boxed{W = F \Delta L}$$

Efficiency

$$\boxed{\eta = \frac{F \Delta L}{P \Delta V}}$$

Losses come from

- friction
- hysteresis
- rubber elasticity
- leakage

12. Fiber Contact Mechanics

Each crossover experiences

Normal force

$$N$$

Friction

$$\boxed{F_f = \mu N}$$

where

$$(\mu)$$

is the friction coefficient.

This friction causes

- hysteresis
 - heat generation
 - efficiency loss
 - wear
-

13. Torque Theory

Clockwise fibers

produce torque

$$\backslash T_c \backslash$$

Counterclockwise fibers

produce

$$\backslash T_{cc} \backslash$$

Balanced braid

$$\backslash \boxed{ T_c = T_{cc} } \backslash$$

Therefore

$$\backslash \boxed{ T_{net} = 0 } \backslash$$

This explains why a symmetric braid contracts without twisting.

14. Radial Stiffness

The braid behaves like a geometric spring.

Approximate radial stiffness

$$\backslash \boxed{ k_r = \frac{dF_r}{dr} } \backslash$$

Small braid angle

↓

higher radial stiffness

Large angle

↓

lower stiffness

15. Fatigue Theory

Each cycle produces

Pressure

↓

Fiber tension

↓

Contact force

↓

Sliding

↓

Wear

↓

Fatigue

Fatigue life depends on

$$N_f = f(P, \theta, \mu, \sigma_f)$$

where

- pressure
- braid angle
- friction coefficient
- fiber stress

all influence the number of cycles before failure.

16. Design Relationships Summary

Design Variable	Governing Relation	Effect
Pressure	$(F \propto P)$	Increases force
Braid angle	$(F \propto (3 \cos^2 \theta - 1))$	Controls contraction/extension
Fiber length	$(b = \sqrt{L^2 + (2 \pi n r)^2})$	Governs geometry
Fiber stress	$(\sigma = T/A)$	Material selection
Hoop stress	$(\sigma_h = Pr/t)$	Bladder design
Axial stress	$(\sigma_a = Pr/2t)$	End fitting design
Friction	$(F_f = \mu N)$	Hysteresis and wear
Efficiency	$(\eta = \frac{F \Delta L}{P \Delta V})$	Performance
Strain	$(\epsilon = \frac{L - L_0}{L_0})$	Stroke
Torque	$(T_c = T_{cc})$	Prevents twisting

For a research project

If you want to develop a **novel McKibben sheath**, the theoretical framework should combine four areas:

1. **Pressure vessel mechanics** for bladder stresses.
2. **Helical geometry** to relate fiber length, diameter, length, and braid angle.
3. **Force equilibrium** to derive axial force and contraction.
4. **Contact mechanics** to model friction, tension distribution, hysteresis, and fatigue at braid crossovers.

Most published work uses only the first three. Incorporating detailed contact mechanics and fiber interaction into the design model is an area where there is still room for meaningful research.

16. Design Summary

This creates a copy that others can chat with



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